

R&D / NLP PROTOTYPE

Recruiter–Candidate Call Analytics

Automated checklist verification and speech-pattern signals of
call quality

Portfolio project · Python · Google Colab



01



Did the recruiter cover the required checklist?

The notebook analyzes diarized recruiter–candidate phone calls: it checks whether a required set of topics was covered, computes a compliance score against that checklist, and explores speech-pattern signals that may correlate with call quality.

- Greeting
- Citizenship
- Age
- Health
- Salary
- Paperwork

From transcribed CSV to per-utterance rows

- 1 CSV export of calls: the *transcription* field holds a JSON payload of diarized, timestamped utterances with emotion tags.
- 2 *json_normalize + explode* — the nested list of utterances is flattened: one dataframe row per utterance.
- 3 A second *normalize + concat* pass expands nested result fields into flat columns.
- 4 Split by *spk* — separate subsets for the recruiter (*spk* == 0) and the candidate (*spk* == 1).

Technology Stack

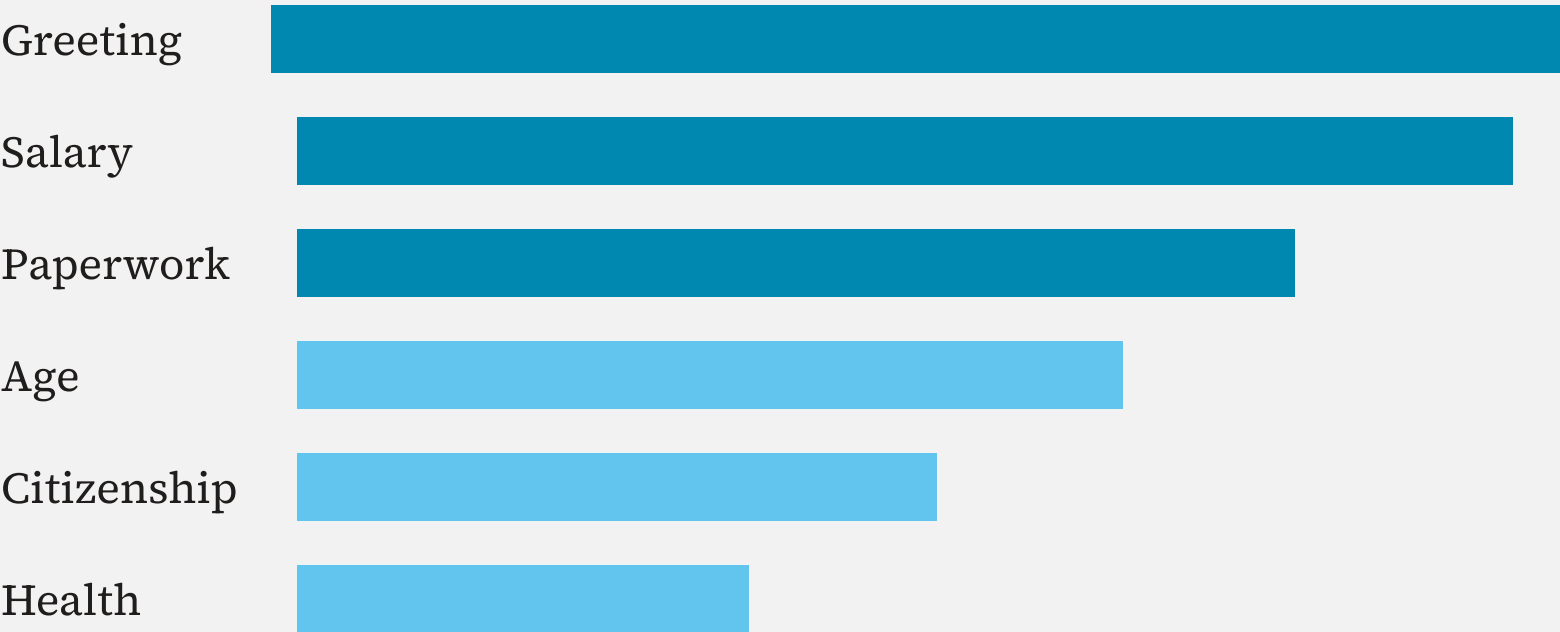
COMPONENT	TECHNOLOGY	PURPOSE
Environment	Python 3, Google Colab	Notebook development and execution
Data	pandas, numpy, json	Tabular processing, JSON parsing
Tokenization	NLTK (punkt, stopwords)	Tokens, noise reduction
Lemmatization / stemming	pymorphy3, SnowballStemmer, custom Porter (RU)	Normal and root word form
Keyword search	re (regular expressions)	Rule-based topic detection
Statistics	phik	Correlation across mixed-type features
Visualization	matplotlib, seaborn, plotly, wordcloud	Charts, heatmaps, word clouds

A checklist compliance score

0-10

A keyword/regex classifier tags each recruiter utterance with a checklist category. The score equals the share of on-topic utterances out of all utterances.

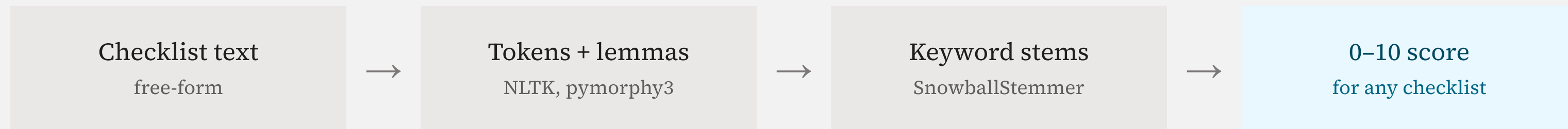
CHECKLIST CATEGORY DISTRIBUTION



Illustrative mockup · normalized histogram (Plotly)

From one fixed checklist to any custom text

The rule-based classifier only works for one fixed checklist. The generalized engine turns a one-off script into a reusable scoring approach.



A Single Reusable Preprocessing Function

- 01 Lowercasing and Russian `ë` → `e` normalization
- 02 Regex cleanup: HTML tags, punctuation and ellipses, repeated whitespace, encoding artifacts
- 03 Tokenizing utterances into words
- 04 Lemmatization — `pymorphy3` and `NLTK WordNetLemmatizer`
- 05 Filtering stopwords and non-alphabetic tokens

Who Talks, and How

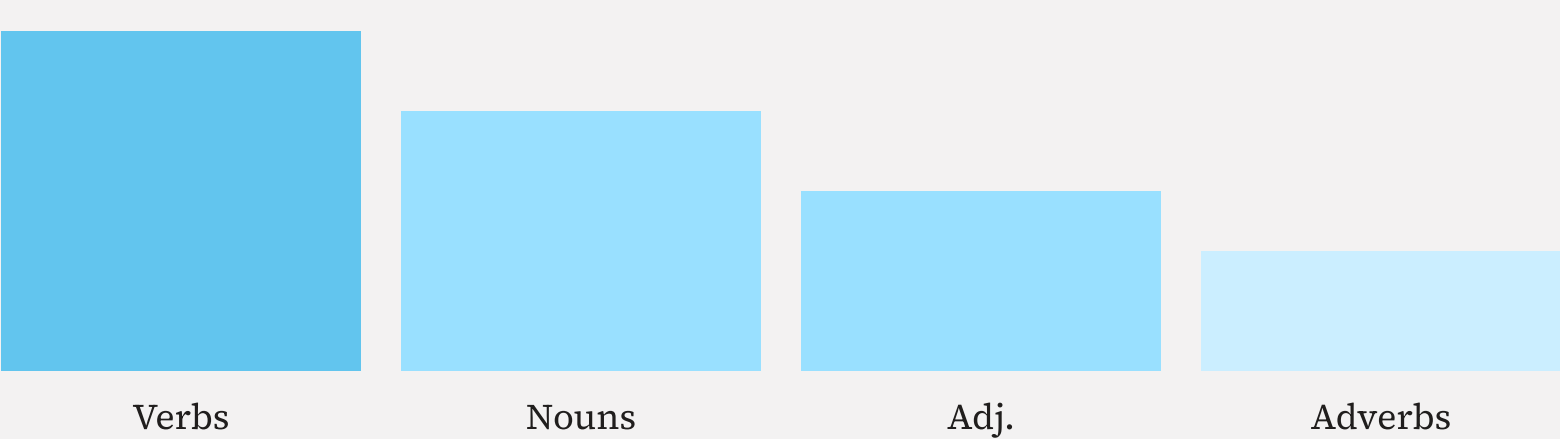
Talk-time ratio



Hypothesis: healthy calls fall near a target ratio close to 60/40.

Illustrative mockup

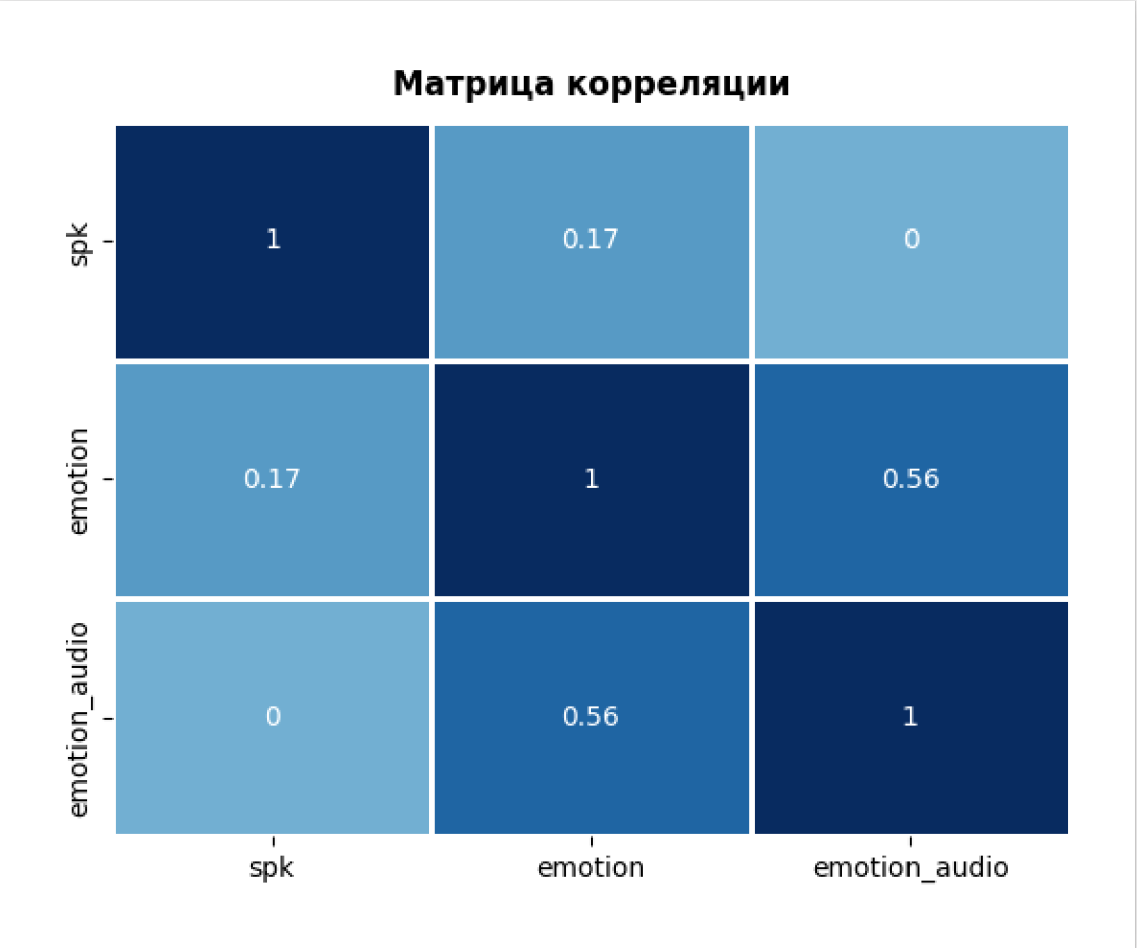
Recruiter part-of-speech profile



Hypothesis: a higher verb ratio correlates with more persuasive, active speech.

Do Text and Audio Emotion Agree?

A *phik* correlation matrix between speaker, text-derived emotion, and audio-derived emotion tests whether the two signals agree and whether either correlates with speaker role.



Notebook output · phik correlation

Dominant Vocabulary, by Role



Recruiter



Candidate

An Exploratory Prototype, Not a Production System

The notebook demonstrates a methodology: rule-based and keyword-based NLP for Russian text, a custom call-quality scoring metric, statistical correlation analysis, and data visualization.

*All company-, candidate-, and vacancy-specific details have been generalized or omitted
— only the technical methodology is documented.*

